Active Directory

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Active Directory

– To understand **Active Directory (AD)** and its role in a network, you need to know what a **directory service**.

Directory Service

- A network **directory service (DS),** Is a service that
  - **stores** information about a computer network.
  - **Offers** features for retrieving and managing that information.

- AD is a **database** composed of **records** or **objects** describing
  - Users.
  - available network resources, such as servers, printers, and applications.

Directory Service (cont.)

- **Directory service** includes functions to:
  - search for, add, modify, and delete information.

- **Directory service** can also:
  - manage how resources can be used and by whom.

Windows Active Directory

- **Active Directory**
  - is the core component in a Windows domain environment

- **Active Directory**
  - is a **directory service** based on standards for defining, storing, and accessing directory service objects.

Windows AD Protocol

- Lightweight Directory Access Protocol (LDAP)
  - X.500 Standard.
  - Based on TCP/IP
  - A method for accessing, searching, and modifying a directory service.
  - A client-server model.

Active Directory Structure

There are two aspects of Active Directory's structure:

– Physical structure
– Logical structure.

Active Directory's Physical Structure

- The physical structure of AD consists of sites and servers configured as domain controllers.
- **site** is a physical location.
- Microsoft defines a site as one or more IP subnets connected by high-speed LAN technology.

Active Directory's Physical Structure (cont.)

– A **domain controller (DC),**
  - which is a computer running Windows Server with the Active Directory Domain Services role installed.

Active Directory's Logical Structure

There are **four** organizing components of Active Directory:

- Organizational units (OU)
- Domains
- Trees
- Forests

Active Directory's Logical Structure (cont.)

- An **organizational unit (OU)**
  - is used to organize a network's users and resources into logical administrative units.
  - It often mimics a company's internal administrative structure,
- An OU contains Active Directory objects, such as
  - user accounts, groups, computer accounts, printers, shared folders, applications, servers, and domain controllers.

Active Directory's Logical Structure (cont.)

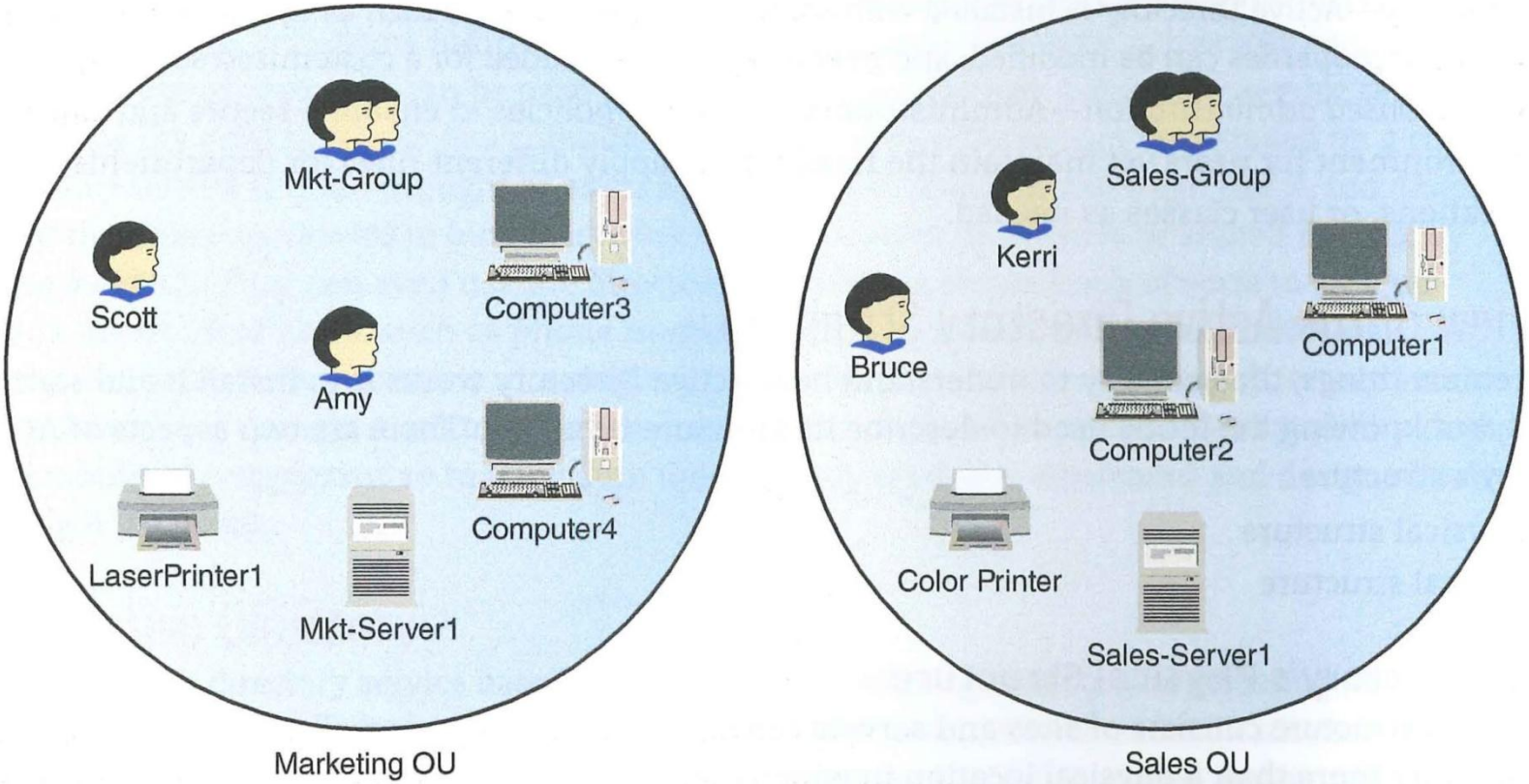

Figure 1-1 Active Directory organizational units

Active Directory's Logical Structure (cont.)

– A **domain** is:

- Active Directory's core structural unit. It contains OUs and represents administrative, security, and policy boundaries

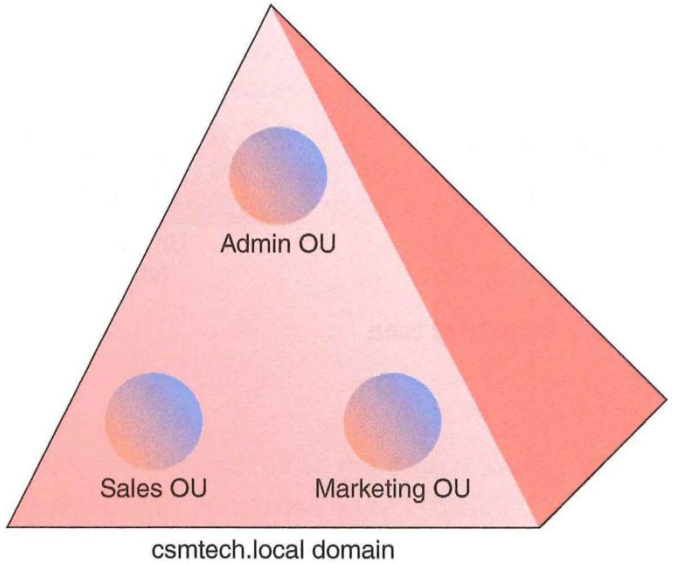

Figure 1-2 An Active Directory domain and OUs

Active Directory's Logical Structure (cont.)

- An AD **tree** is a **grouping** of domains that share a common naming structure.

- A **tree** consists of a **parent** domain and possibly one or more **child domains** (also called *subdomains)*.

Activ icture

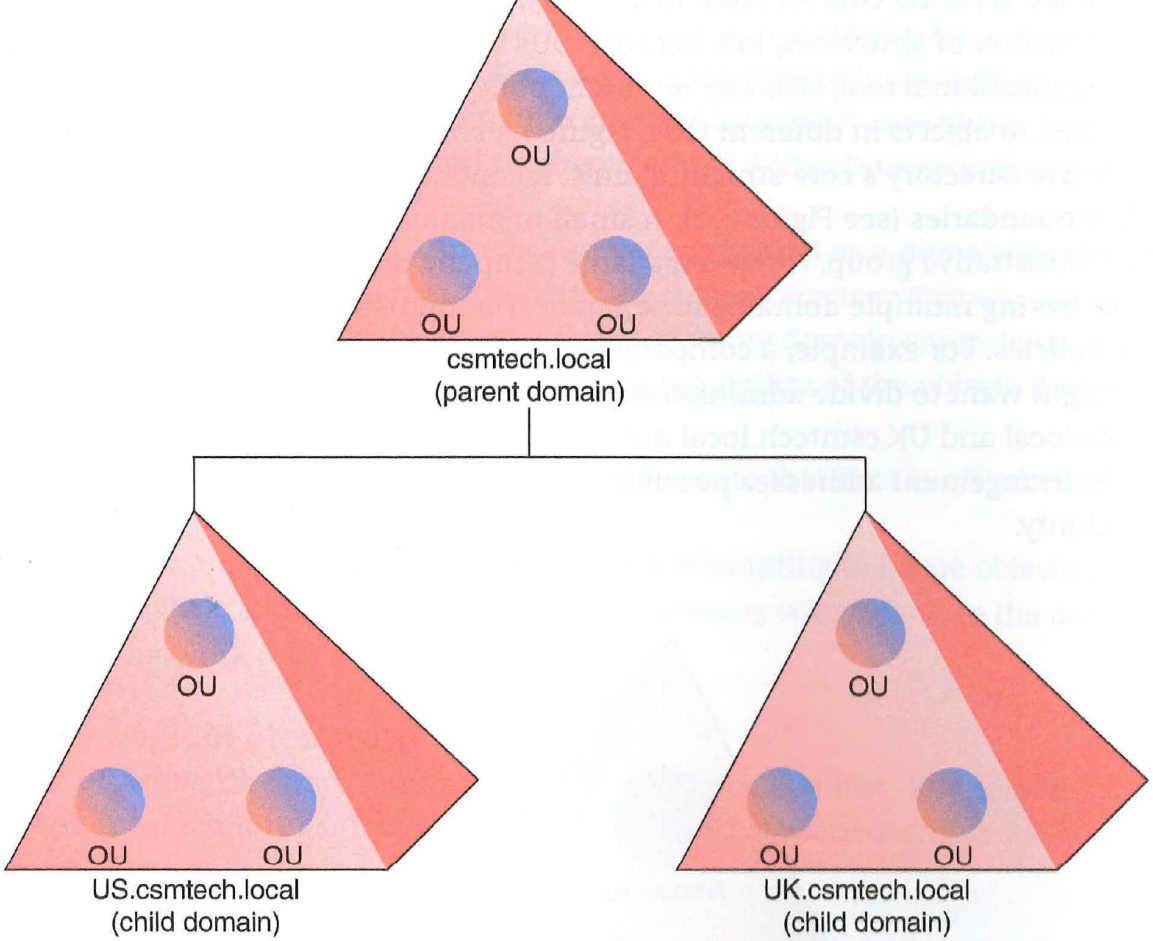

Figure 1-3 An Active Directory tree

Active Directory's Logical Structure (cont.)

- An AD **forest** is a collection of one or more trees.
- A forest can consist of a single tree with a single domain, or it can contain several trees, each with a hierarchy of parent and child domains.
- Each tree in a forest has a different naming structure

Active Directory's Logical Structure (cont.)

A **forest's main purpose** is to provide a common Active Directory environment in which:

- all **domains** in all trees can **communicate** with one another and
- share information to allow **independent** operation and administration of each domain.

Active Directory's Logical Structure (cont.)

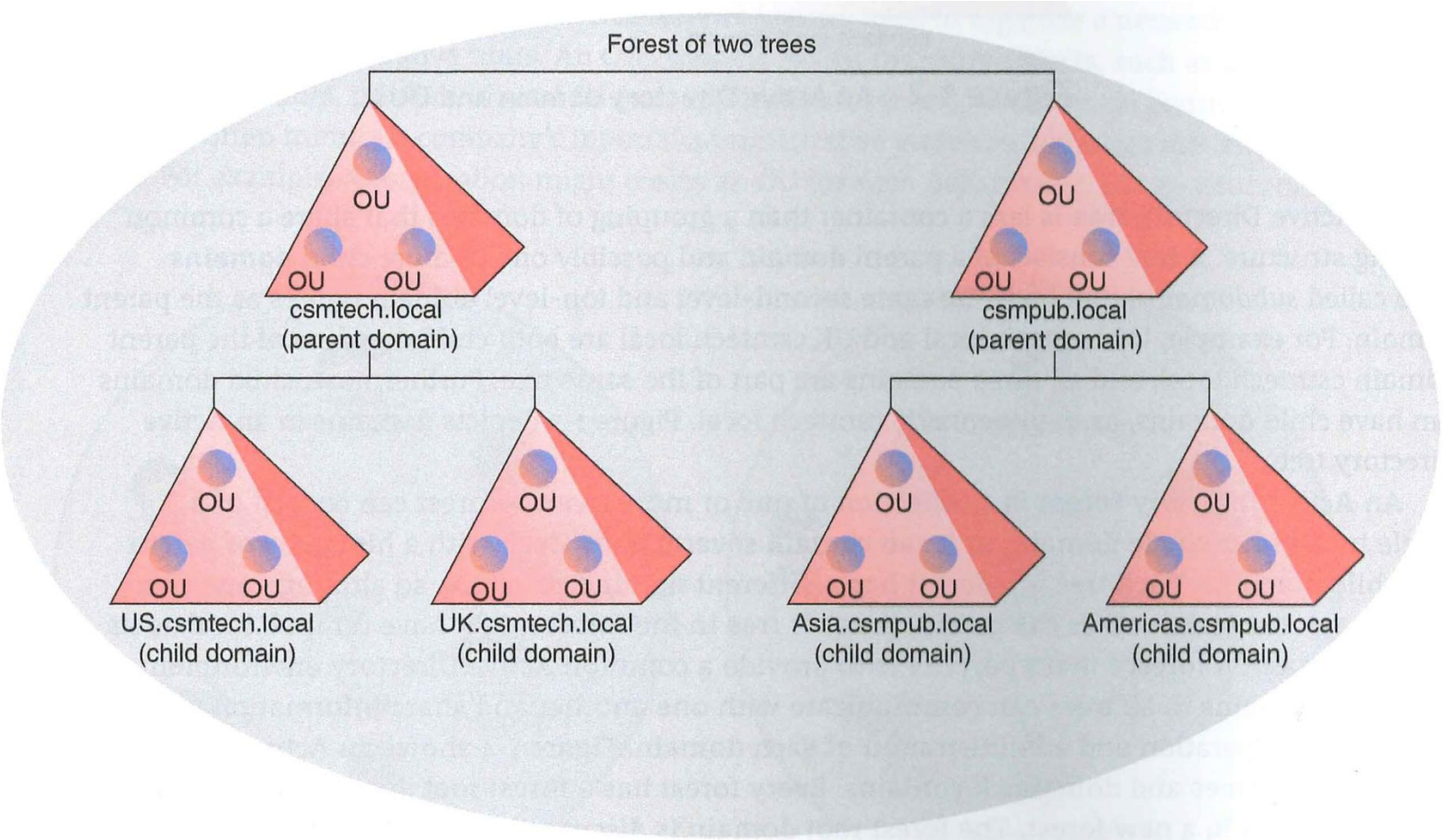

Figure 1-4 An Active Directory forest

Discussion and Questions